Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 34)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 State the Brundtland Commission (1987) definition of sustainable development.

[1 mark]

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0 1 . 2 Explain why activities such as slash-and-burn agriculture, fishing and sewage discharge can be sustainable with a small human population but not with a large one.

[3 marks]

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0 1 . 3 Explain why urban populations are often less aware of unsustainable exploitation than rural communities.

[2 marks]

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0 1 . 4 Describe **two** negative feedback mechanisms in natural systems and explain how human responses can override them.

[4 marks]

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0 2 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

0 2 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

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0 3 . 1 Explain how natural material cycles differ from linear human systems of material use.

[4 marks]

0 3 . 2 Explain why the use of toxic metals in electronic products is a particular problem for sustainability.

[2 marks]

0 3 . 3 Describe how the design of a product can move it from a linear to a cyclical system.

[4 marks]

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0 4 A new lowland nature reserve is being created on former farmland next to an existing ancient woodland.

0 4 . 1 Explain why the **area** and the **shape** of the reserve both matter in its design.

[3 marks]

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0 4 . 2 Explain the value of a biological corridor linking the new reserve to the ancient woodland.

[2 marks]

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0 4 . 3 Explain how each of the following abiotic factors could be managed to increase the biodiversity of the reserve.

[4 marks]

Water levels .....

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Soil nutrient levels .....

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Topography .....

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Light levels .....

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0 4 . 4 Much of the reserve will be kept as a **plagioclimax**. Explain what this means and how grazing is used to maintain one.

[3 marks]

0 4 . 5 Explain why the removal or culling of some species may be necessary on the reserve.

[3 marks]

15

0 5 . 1 Describe how agriculture alters the hydrological cycle.

[3 marks]

Do not write outside the box

0 5 . 2 Explain how each of the following consumer choices can reduce the environmental impact of food production.

[4 marks]

Seasonal / local food .....

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Organic food .....

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0 5 . 3 Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

[3 marks]

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0 6 Table 1 shows the number of bird species and butterfly species recorded in three woodlands in the same region.

Table 1

Woodland	Management	Bird species	Butterfly species
A	Ancient oak wood, coppiced on rotation with clearings	42	23
B	Ancient oak wood, unmanaged for 60 years, dense canopy	28	9
C	40-year-old conifer plantation	14	3

0 6 . 1 Use Table 1 and your knowledge to explain the differences in biodiversity between the three woodlands.

[5 marks]

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0 6 . 2 Describe **three** conservation efforts used to protect UK woodlands.

[3 marks]

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0 6 . 3 Explain why a high biodiversity increases the ecological stability of a woodland.

[2 marks]

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0 7 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 7 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 7 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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0 8 . 1 Complete Table 1 by naming the most suitable statistical test for each hypothesis.

[4 marks]

Table 1

Hypothesis	Statistical test	
Noise levels decrease as distance from a road increases		
Fewer cars use a road after road charging is introduced (counts on 12 days before and after)		
Woodlands managed in different ways have different numbers of dormice		
Mean lobster mass is higher in areas where collection is banned		

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0 8 . 2 Explain why the chi-squared test cannot be used to compare the dissolved oxygen concentrations of two rivers.

[2 marks]

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0 8 . 3 Explain why a scientific investigation never 'proves' a hypothesis.

[2 marks]

0 8 . 4 Describe how error bars are added to a graph of means and how they are interpreted.

[2 marks]

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0 9 . 1 Explain what is meant by **biomimetics** and describe **three** examples.

[4 marks]

0 9 . 2 Explain why many plants produce chemicals that are useful as human medicines, and give **two** examples.

[3 marks]

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0 9 . 3 Explain why the discovery of medicines is a strong argument for conserving whole habitats rather than individual species.

[2 marks]

0 9 . 4 Give **one** example of a wildlife species used in physiological research and the human health issue it helps to understand.

[1 mark]

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1 0 . 1 Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

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Extra space

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1 0 . 2 State what is meant by a **fed** aquaculture species.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

OR

1 1 . 2 Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS